



EXPLORE FLIGHT

WE'RE WITH YOU WHEN YOU FLY

Provider of Services to UAM

An overview and initial use case discussion

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18 May and 15 June 2021

Outline

PSU overview with ConOps tour

Two PSU workflows

Definitions and Standards

Use case assumptions

Quiz

Reverse Quiz

Use Case and Discussion (potentially delayed)



The Provider of Services for UAM (PSU) is a key component of the UAM ecosystem, described in multiple ConOps documents.



As we advance through UMLs 1-4, we need insight and input into how the management of the airspace will work.



Federated management of the airspace is a novel concept invented via NASA research and has been validated via multiple means, but open questions remain.



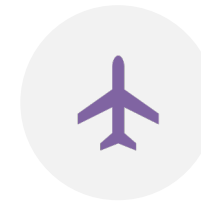
Overview



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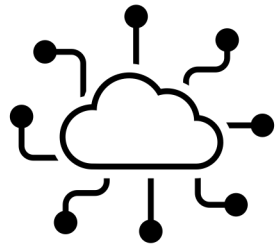
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Federated Airspace Management

Federated enterprise architecture is a collective set of architectures with the following attributes:

- It operates collaboratively, where governance is divided between a central authority and constituent units, balancing organizational autonomy with enterprise needs.
- The central authority's architecture can focus on the dynamics of economies of scale, standards, and the well-being of the enterprise.
- Constituent units' architectures have the flexibility to pursue autonomous strategies and independent processes



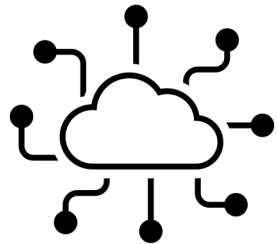


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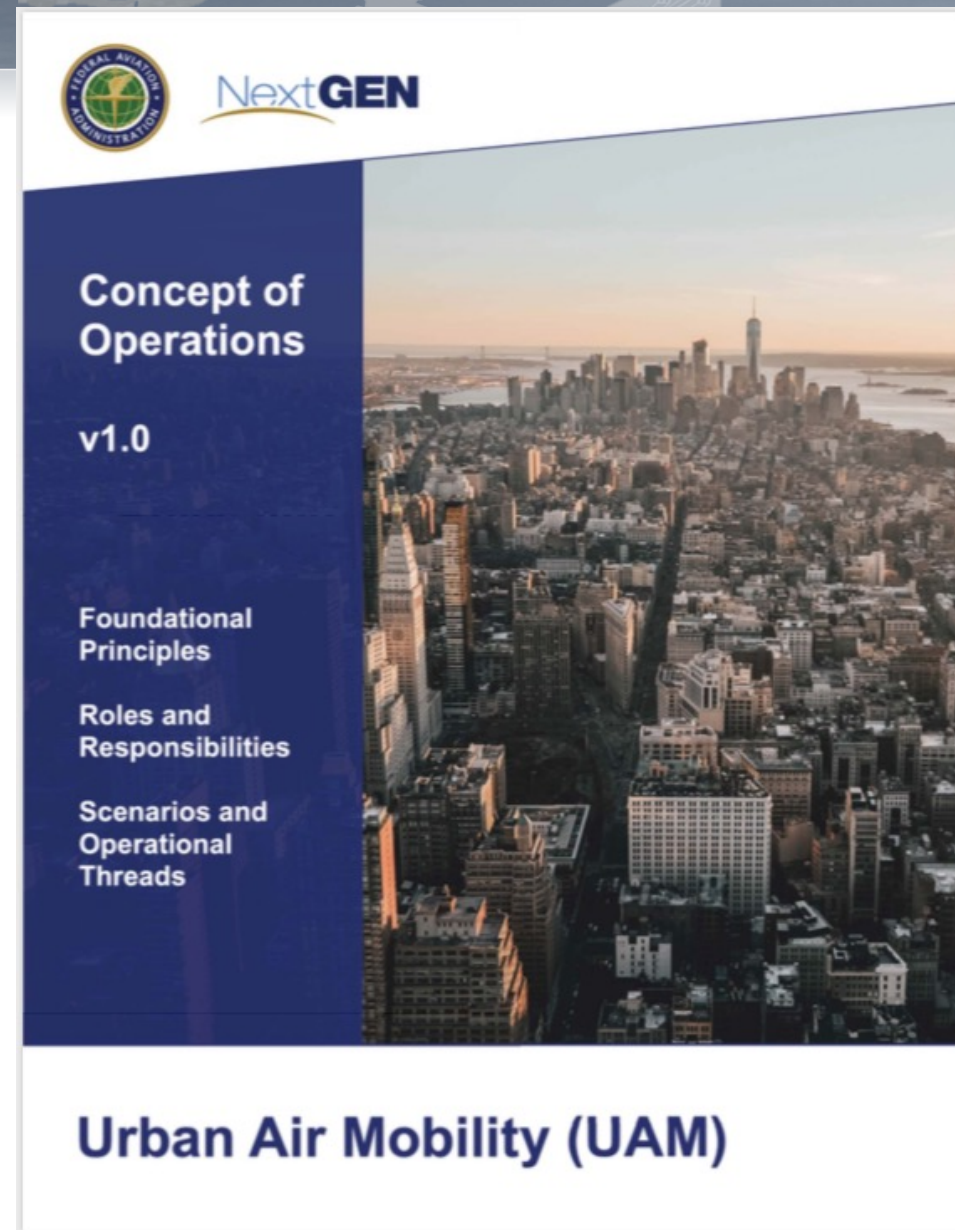




FAA UAM ConOps v1.0

From NextGen Office, published June 2020

https://nari.arc.nasa.gov/sites/default/files/attachments/UAM_ConOps_v1.0.pdf





2 Overarching Principles and Assumptions

The following principles and assumptions will be used when guiding the development of the UAM operating environment and evolving the concept:

...

- Providers of Services for UAM (PSUs) will be utilized by operators to receive/exchange information during UAM operations.
- PSUs will be able to obtain UTM flight information via the UAS Service Supplier (USS) network, and the USS network will be able to obtain UAM flight information via the PSU network.
- UAM operators maintain conformance to shared intent; operators, via PSUs, are aware of intent of other operations in the vicinity.

...

4.3.4 Provider of Services for UAM (PSU)

A PSU is an entity that supports UAM operators with meeting UAM operational requirements that enable safe, efficient, and secure use of the airspace. A PSU:

1. Provides a communication bridge between federated UAM actors, PSU to PSU via the PSU Network, to support the UAM operator's ability to meet the regulatory and operational requirements for UAM operations
2. Provides the UAM operator with information gathered from the PSU Network, about planned UAM operations in a UAM Corridor so that UAM operators can ascertain the ability to conduct safe and efficient missions
3. Analyzes and confirms that a UAM operator's Operational Intent is complete, consistent with current advisories and restrictions, and strategically deconflicted considering already established flight intent of other UAM operators, the CBRs, UAM Corridor capacity, airspace restrictions, UAM aerodrome resource availability, and adverse environmental conditions
4. Provides the confirmed flight intent to the PSU network
5. Distributes notifications (e.g., constraints, restrictions) for the intended area of operation
6. Distributes FAA operational data and advisories, weather, and supplemental data
7. Supports cooperative separation management services (e.g., conformance monitoring, advisory services)
8. Determines UAM Corridor use status
 - a. UAM Corridor use status (e.g., active, inactive) is an indication that one or more UAM operations are occurring in the UAM Corridor
9. Archives operational data in historical databases for analytics, regulatory, and UAM operator accountability purposes

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Joey's Groupings:

PSU-PSU and PSU-
Operator Comms

ATM-like services (but
not ALL ATM-like
services)

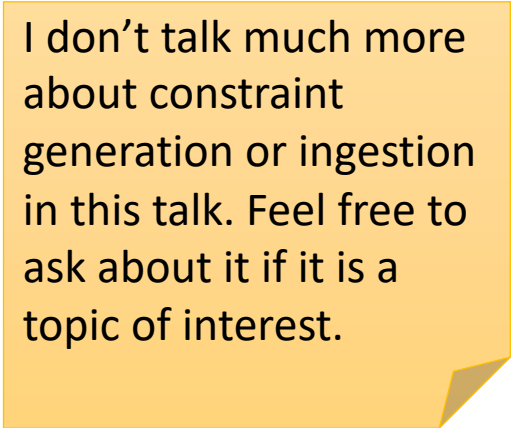
Compliance and audit

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I don't talk much more about constraint generation or ingestion in this talk. Feel free to ask about it if it is a topic of interest.



PSUs facilitate the primary layer of deconfliction.

"Primary" = "first"

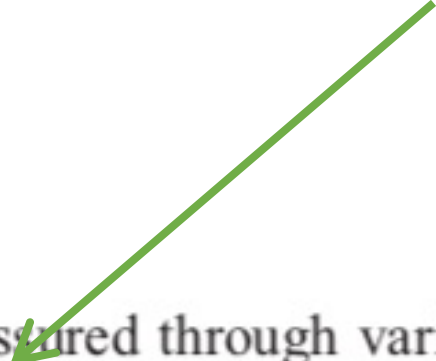
"Primary" != "only"

"Primary" != "most safety critical"

"Primary" != "most important"

4.5 UAM Separation

Separation of operations within UAM Corridors is assured through various strategic and tactical methods. The primary method is strategic deconfliction based on collaborative flight intent sharing. Tactical separation is allocated to the UAM operators, including PIC and aircraft capabilities, and may include support from the PSUs.



NASA UAM Vision ConOps

As summarized in “An Initial Concept for Intermediate-State, Passenger-Carrying Urban Air Mobility Operations”

<https://arc.aiaa.org/doi/10.2514/6.2021-1626>

PSU Excerpts from NASA ConOps

- High-density routes are dynamic based on demand and are managed by PSUs according to consensus standards that are approved by the FAA.
- The FAA retains authority over the UOE and delegates service provision to qualified UOE users (i.e., PSUs), while ensuring the rules and procedures governing UAM operations support safe and secure UAM operations as well as equitable airspace access.
- Specific roles of PSUs and SDSPs with respect to provision of UAM services is defined by FAA qualification requirements for safety-critical services (e.g., surveillance or separation services) and by CBRs* for an area. For example, at UML-4, PSU certification requirements will likely require that PSUs support flight planning by providing strategic conflict detection of proposed operations through sharing of operational intent.

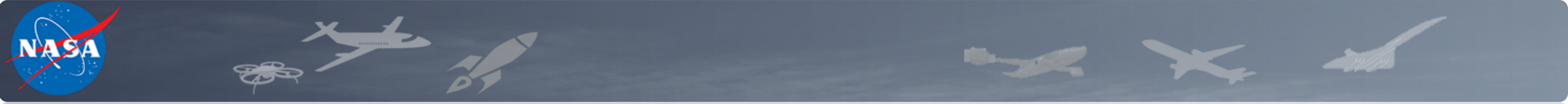
**“Community-Based Rules” are an open discussion item and will evolve in definition and name*

Other ConOps

Industry ConOps generally align the federated airspace management approach.

U-Space in Europe is looking at extending their initial ConOps to include UAM operations.

FAA is developing version 2 of their ConOps with input from more internal stakeholders and using feedback from version 1

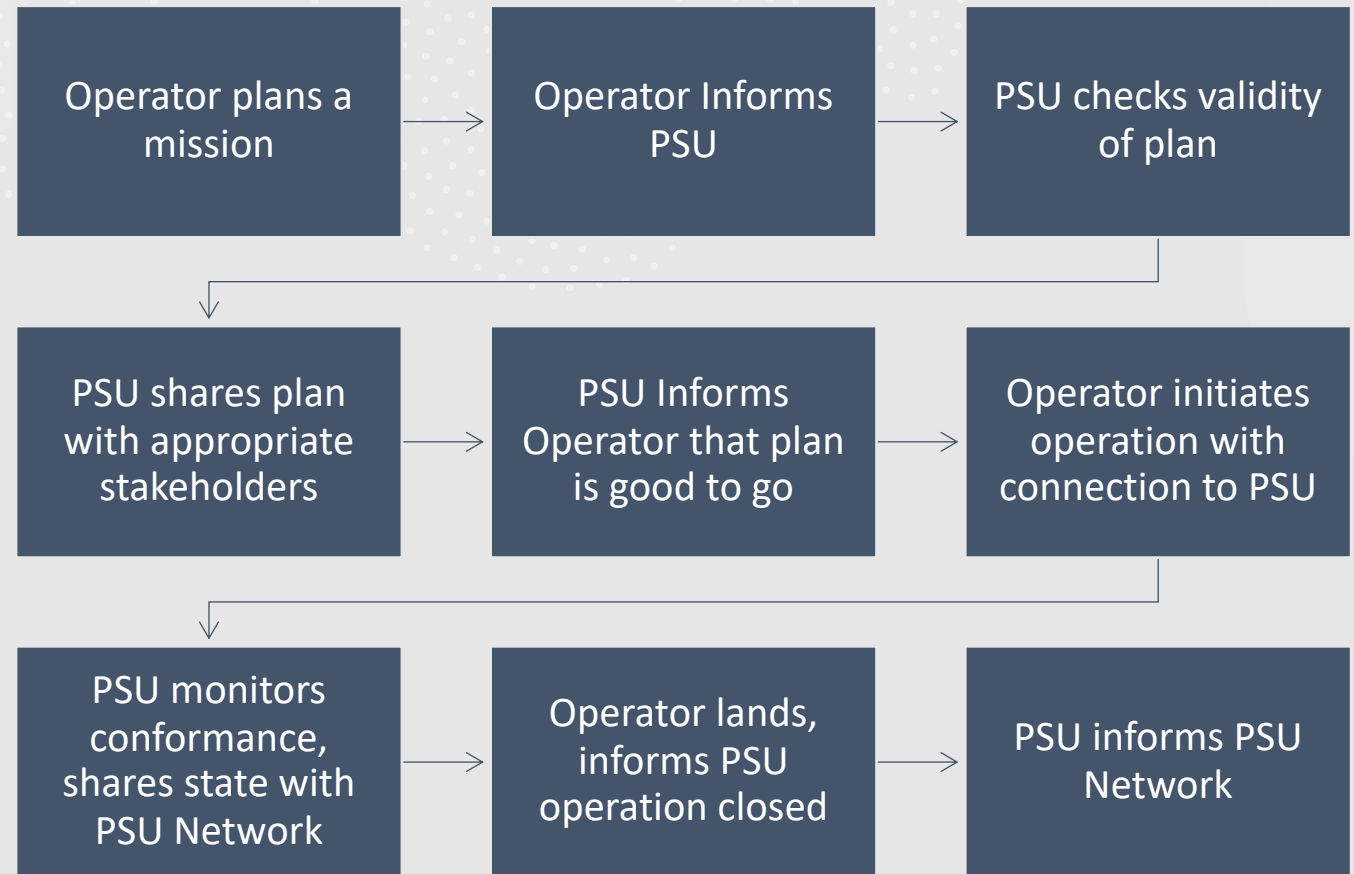


OK... so how does a PSU *actually* work*?

*according to Joey

Nominal UAM Operation

Operator-to-
PSU focused



This is a simplified flow. There are interactions between PSU and FAA not described, there are also several precursor steps to qualify the entities. This simplified flow also excludes vertiport interactions, which are discussed via the use cases to follow. PSU Network interactions also more detailed in subsequent flow.

Nominal UAM Operation

PSU
Network
focused



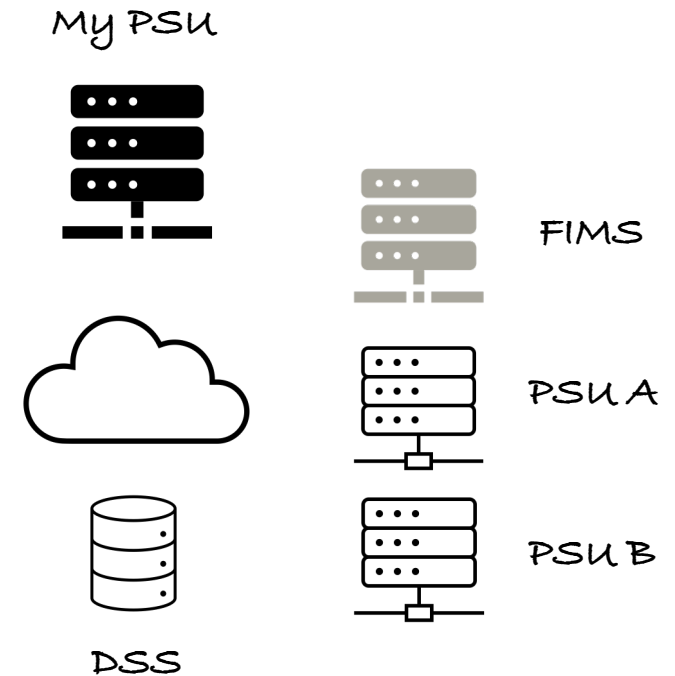
Data and Protocols



1. Operator plans mission
2. Operator submits plan to its supporting PSU
3. PSU checks plan for entities in airspace
4. PSU writes approval to DSS, discovering other PSUs
5. PSU notifies operator of submission success
6. PSU sends approval to other PSUs as required
7. FIMS used throughout and PSU-DSS comms for auth services

Overall example lifted from UTM environment and modified for UAM. While flow is nearly identical, this doesn't imply that data elements or PSU performance requirements are the same.

UTM and UAM are not the same despite some similarities.





Data and Protocols



1. Operator plans mission

2. Operator

sup

3. F

enti

4. F

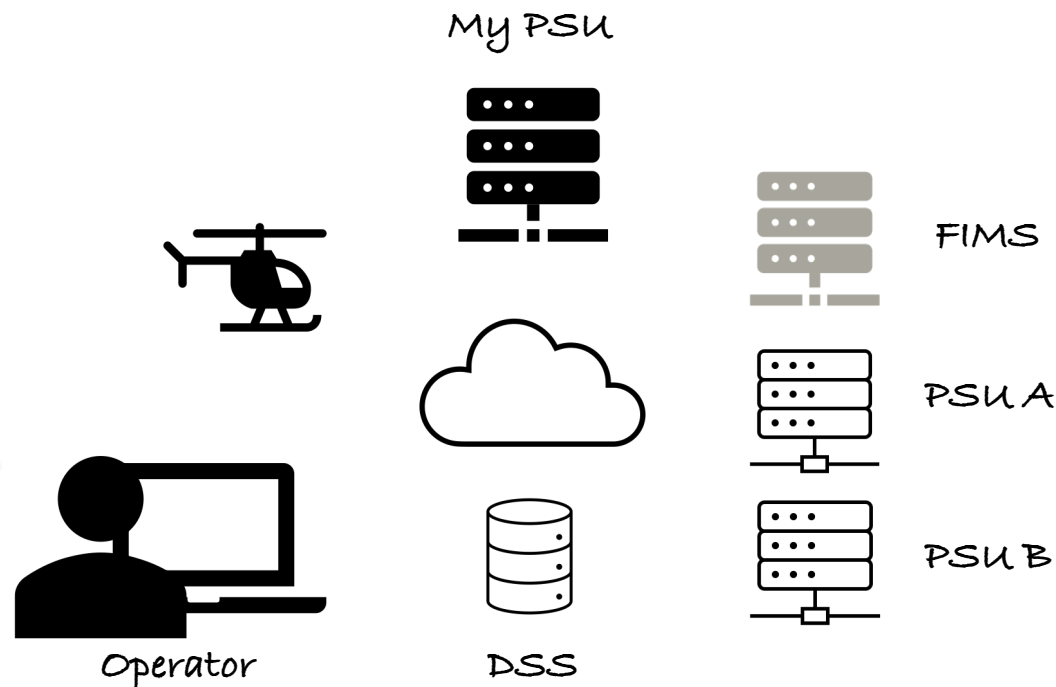
DS

5. PSU notifies operator of plan submission success

6. PSU sends appropriate op data to other PSUs as required

7. FIMS used throughout PSU-PSU and PSU-DSS comms for auth services

Note that the Operator is not necessarily the Pilot. These are distinct roles. The pilot is not included in this simplified interaction, but would have responsibilities.





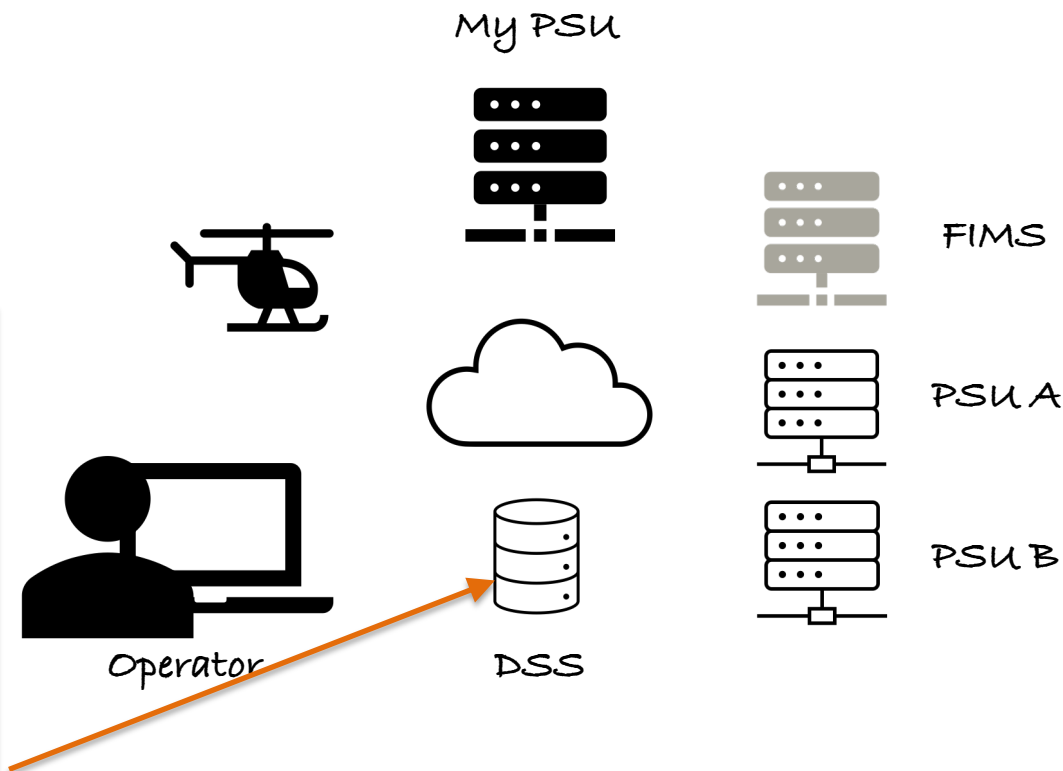
Data and Protocols



1. Operator plans mission
2. Operator submits plan to its supporting PSU
3. PSU checks plan against other entities in airspace
4. PSU writes appropriate op data to DSS, discovering other entities
5. PSU notifies operator of submission success
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7. FIMS used through DSS and PSU-DSS communications

DSS = **Discovery and Synchronization Service**

Industry provided approach to facilitate USS communications in UTM. Assumed here to be leveraged for UAM environment.

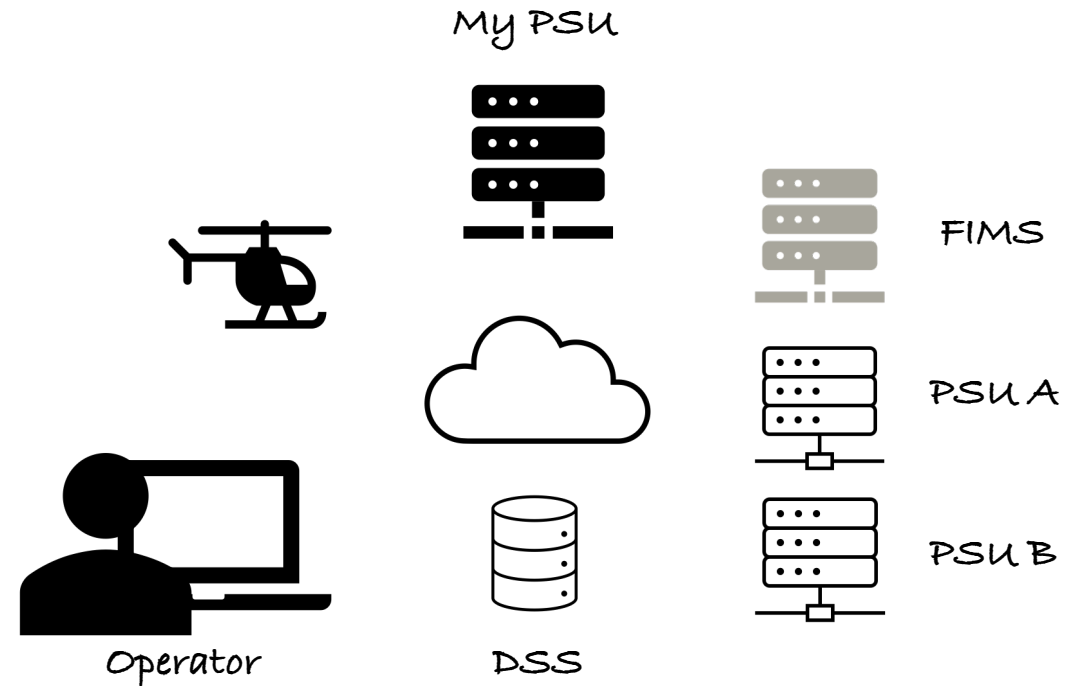




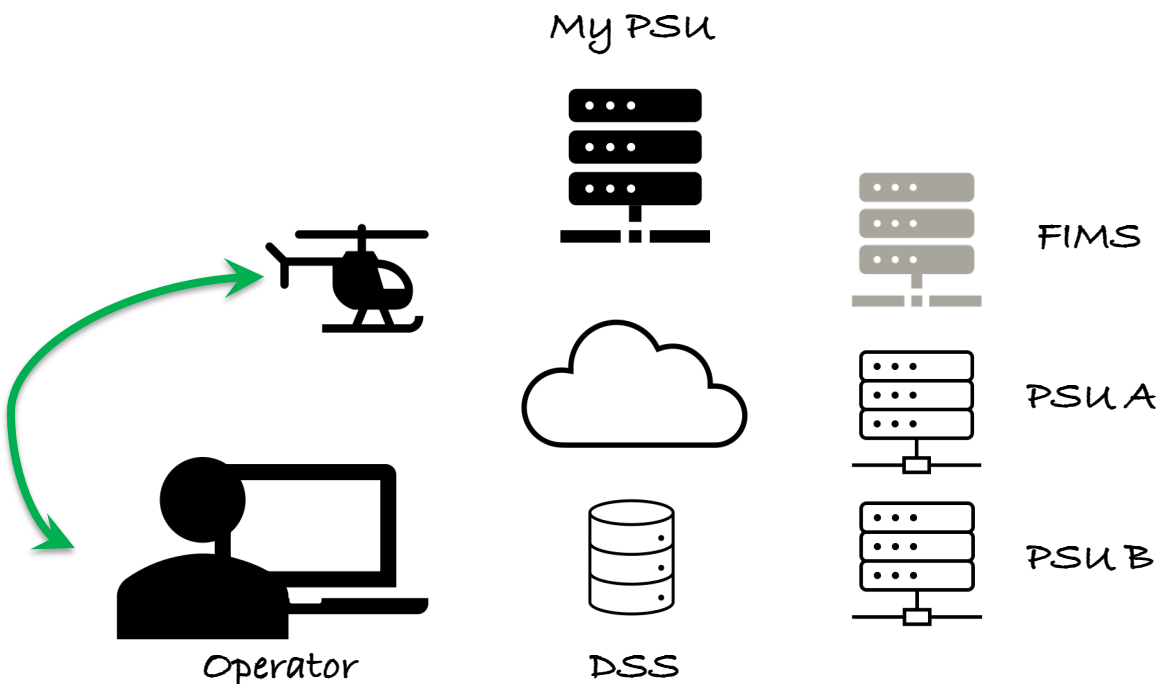
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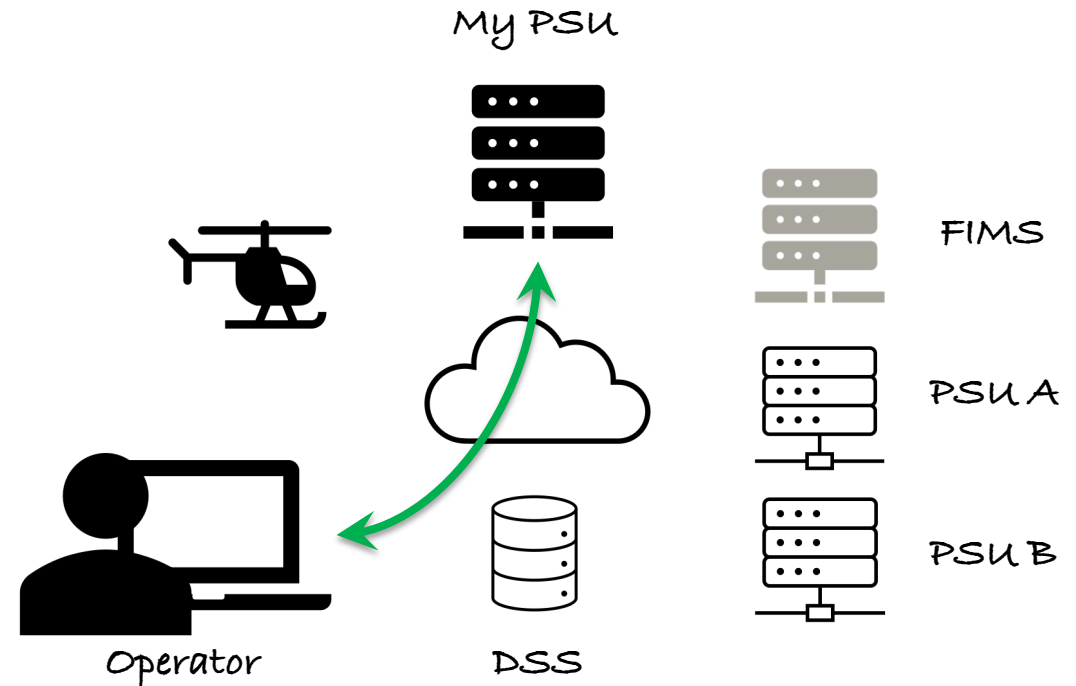




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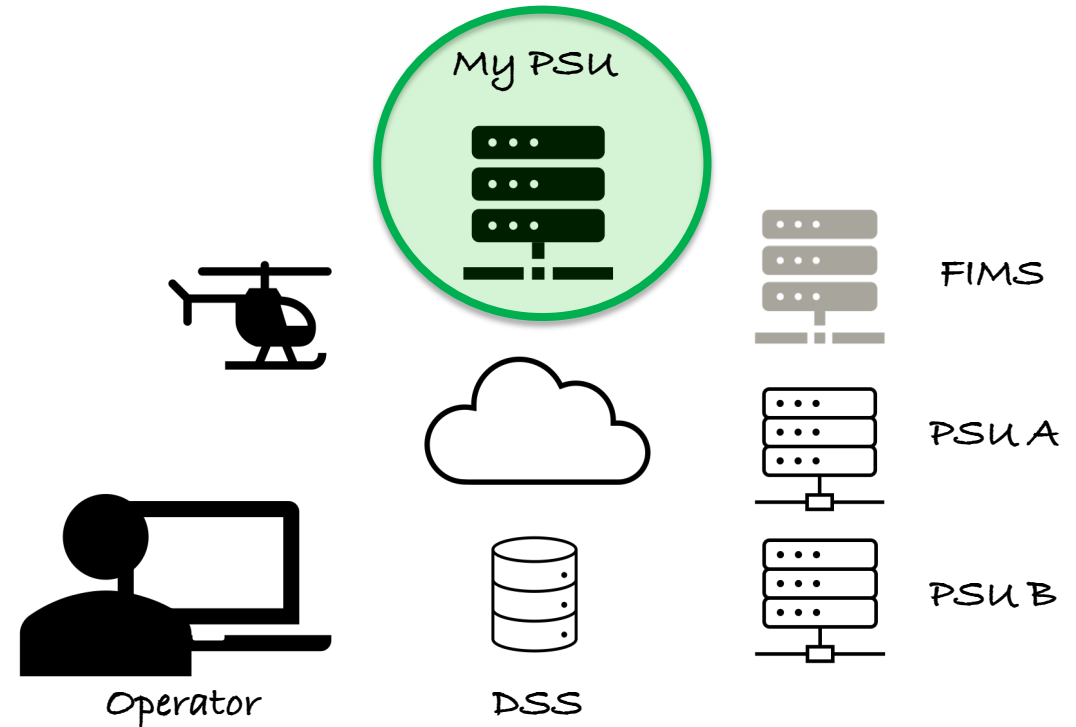




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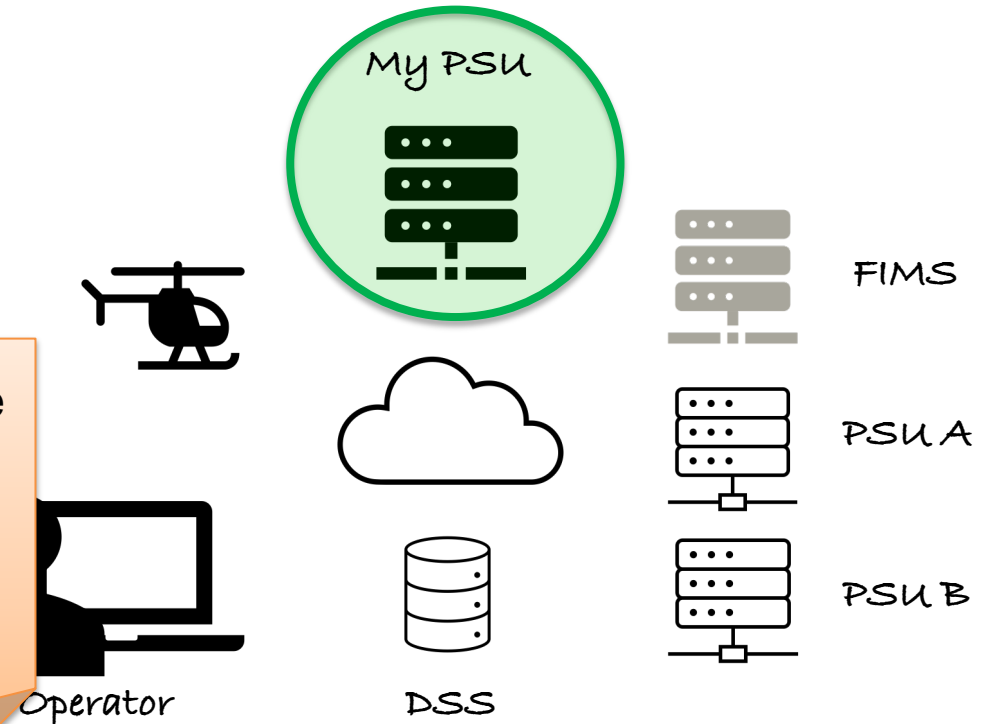




Data and Protocols

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5. PSU notifies operator submission success
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7. FIMS used throughout and PSU-DSS comms for auth services

Also at this stage the PSU may be required to check certain authorizations or declarations in some way. May involve interaction with FIMS or not.

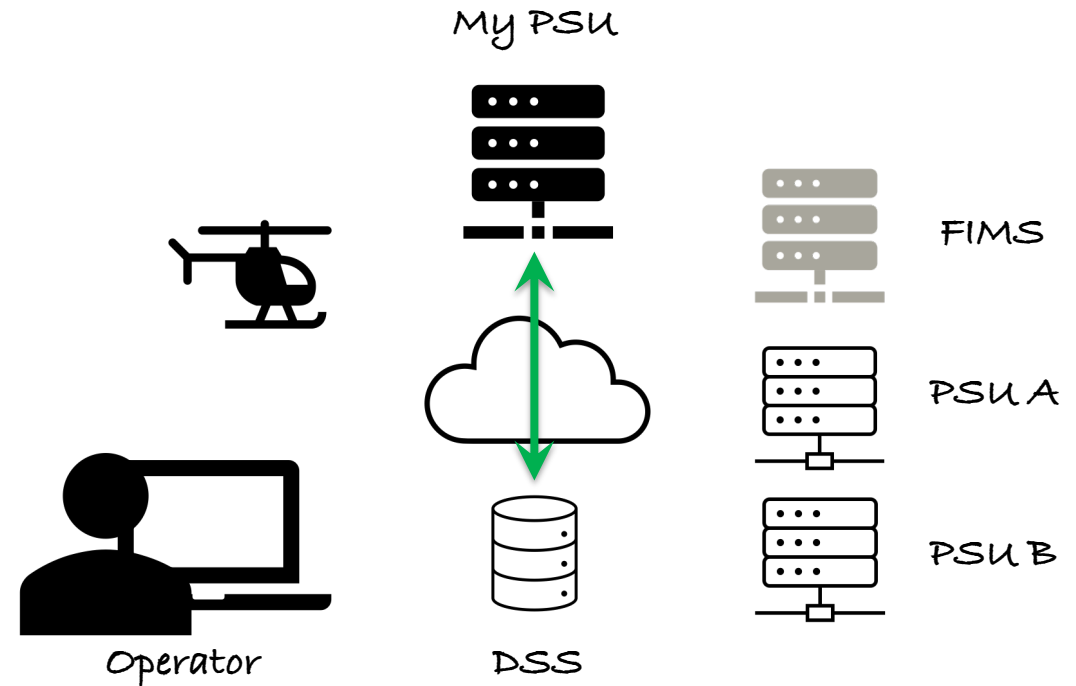




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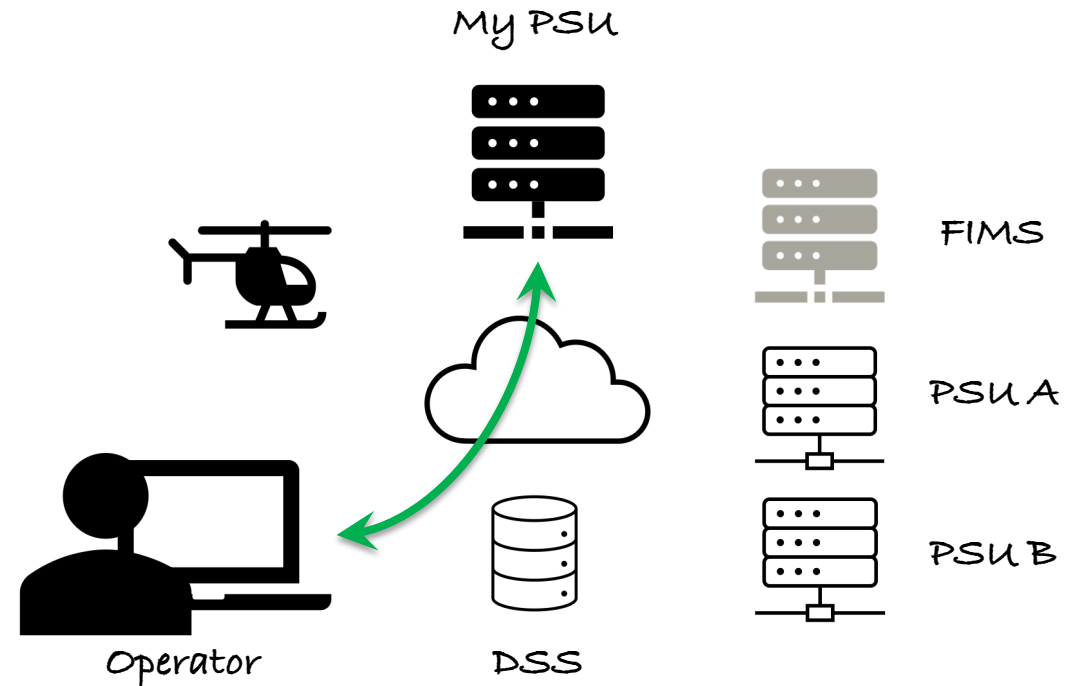




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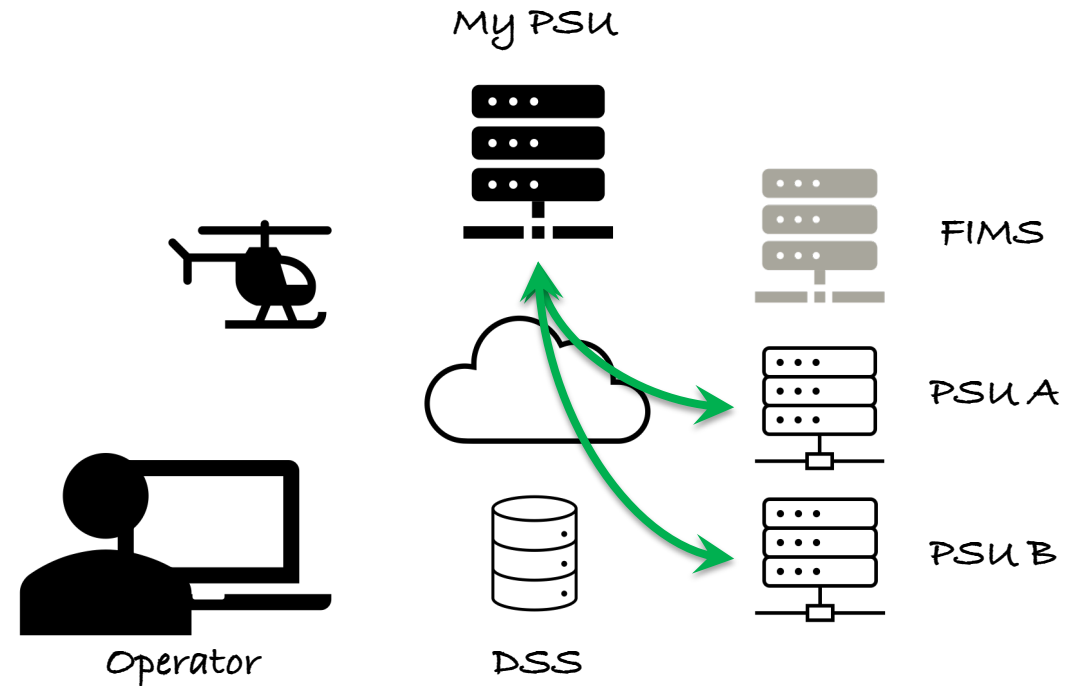




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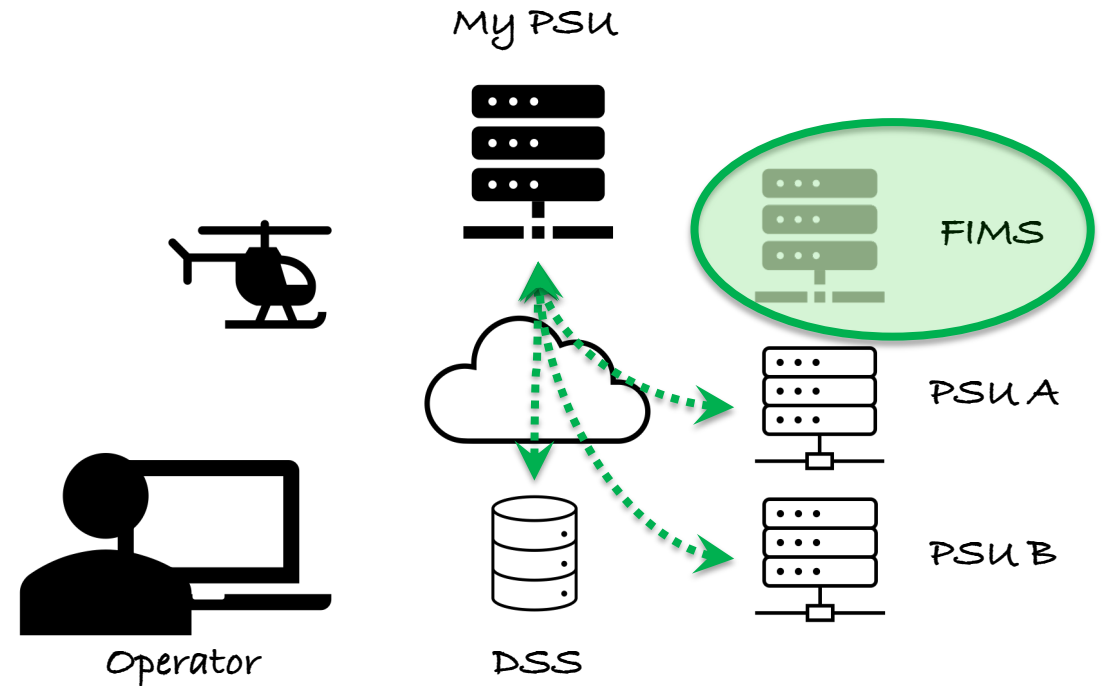




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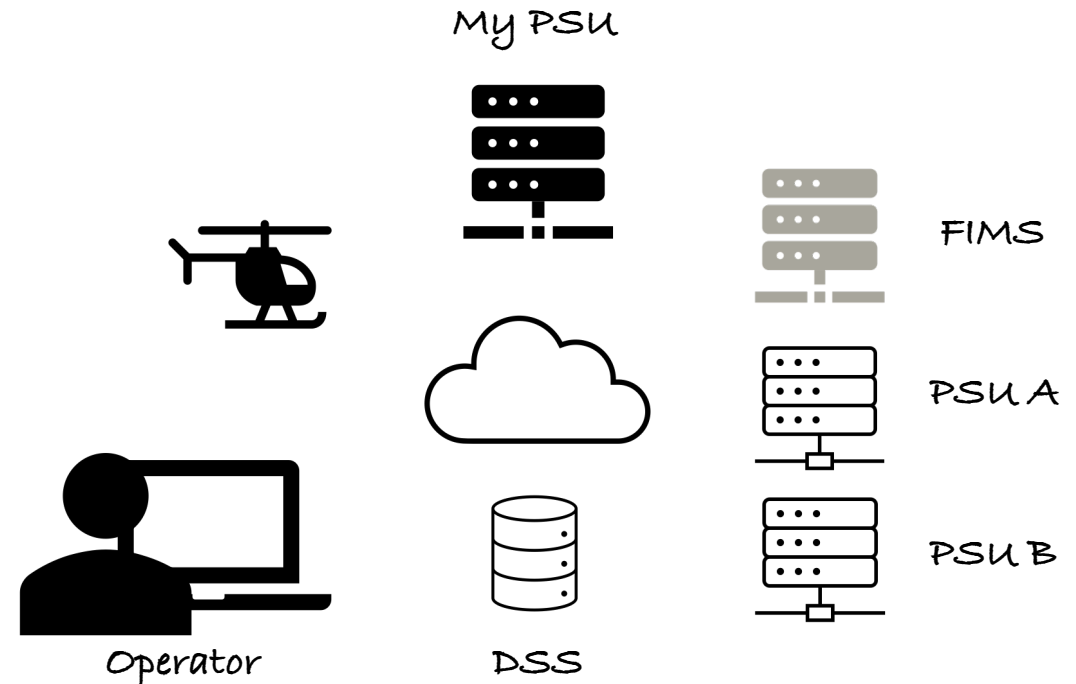




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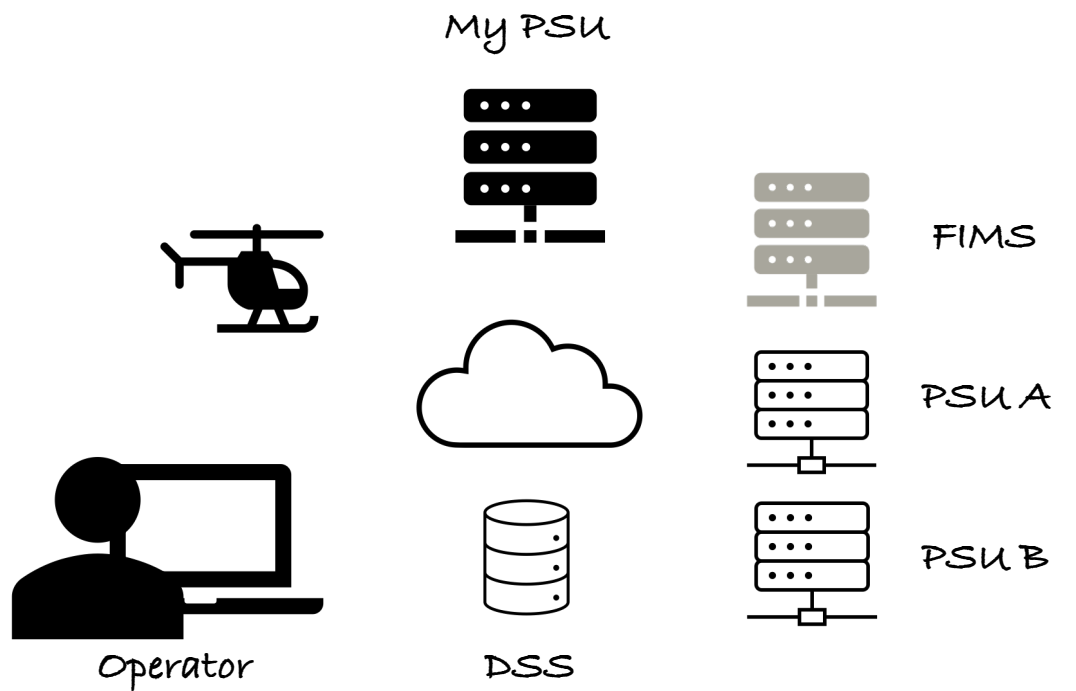
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7. FIMS used throughout PSU-PSU and PSU-DSS comms for auth services



Upon commencing operation, the operator would share state data with its PSU.

Certain changes in the airspace or other operations are pushed to the operator from its supporting PSU. Off-nominal conditions as detected by the PSU or indicated by the operator are pushed to the PSU Network by the supporting PSU. DSS facilitates PSU communications and synchronizing data.

These interactions can be summarized into groups of required and optional services of a PSU. Defining those services is an active research effort with NASA, FAA, and stakeholders.



Non-exhaustive, non-official list. Will evolve with rules and standards.

Other services will be necessary to fully enable UAM operations, but that doesn't mean they are PSU services.

Key PSU services/features



Checking plan versus airspace constructs and accepted rules/protocols

Is plan using airspace constructs appropriately?
Any other required checks performed by operator (wx, etc.)?



Bridging operator-FAA strategic, digital communications

Airspace authorizations
Notification of off-nominal events or encroachment on controlled, non-UAM airspace



Sharing intent and state with other PSUs per CBRs

Ensure that initially shared plans are deconflicted
Provide position/state reports to other PSUs as appropriate



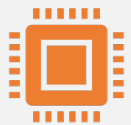
Monitoring conformance of an operation with its plan

Alert operator in case of non-conformance or potentially nearly non-conformant
Update PSU Network in cases of contingency

Provider of Services vs. provider of services



Many entities will likely provide many services in the future UAM and AAM environments



A “PSU” is a specific entity with a yet-to-be-finalized set of minimum capabilities, it is not a generic provider of services. The overall focus of a PSU is on **interoperability** between operators and **bridging** operators and the FAA.



Surveillance is a critical component of UAM and services will be provided by potentially multiple entities... those entities are not necessarily PSUs. Same with weather services, flight plan optimization, battery health monitoring, etc. All important services, but not required to be provided by a PSU and not PSUs simply by virtue of providing those services.

Relation to UTM standards

There are other non-ASTM standards, but these are most relevant to today's discussion.

- ASTM developed a Remote ID standard involving broadcast and networked identification
(<https://www.astm.org/Standards/F3411.htm>)
- ASTM working on a USS Standard including services for:
 - Constraint management (creation and ingestion)
 - Strategic Conflict Detection (intent sharing, conformance monitoring)
 - <https://www.astm.org/DATABASE.CART/WORKITEMS/WK63418.htm>
- Organizations looking to extend the USS standards to cover UAM
- UAM likely to need more initial standardization than small UAS needs
 - Airspace authorizations
 - Continuous network position sharing
 - More robust contingency management

Use Case Assumptions

PLEASE NOTE: These are assumptions for the Use Case... not 'universal' assumptions about UAM.

Airspace construct and procedures exist to facilitate UAM operations.

UAM Operations are in and around DFW.

10's of operations may be airborne at any given time.

Several vertiports are in operation to support UAM.

UAM Operations are capable of VFR flight in uncontrolled airspace.

UAM Operations stay within defined UAM airspace structures when in controlled airspace.

Nothing bad or unexpected happens during the lifespan of our use cases today. Bad stuff is bad.

Which description best fits my current role?

- A. FAA civil servant or contractor
- B. NASA civil servant or contractor
- C. Other government employee
- D. Industry stakeholder for UAM
- E. General Interest

In the environment described via the assumptions, how important (or unimportant) is the network of PSUs supporting these operations?

- A. Not important at all, the environment described can be effectively managed with existing means
- B. Not important because there are other, better ways to manage this airspace that are neither PSU-based nor in existence today.
- C. Somewhat important, PSUs could help with operational awareness and alleviate some burden from today's system, but we might get by without them with some effort.
- D. Quite important, it is unlikely this environment could effectively operate without a PSU Network.
- E. Indispensable. There is no other clear way that this environment could effectively exist without a PSU Network.

When is the earliest time in the life cycle of a UAM operation that ATC needs to be aware of that Operation?

- A. As early as possible or as soon as the PSU is aware of the operator's intent to operate.
- B. Just before takeoff.
- C. Only upon entry to controlled airspace.
- D. Only at the time of some off-nominal condition
- E. Never

How should nominal communications occur between a UAM operator and ATC?

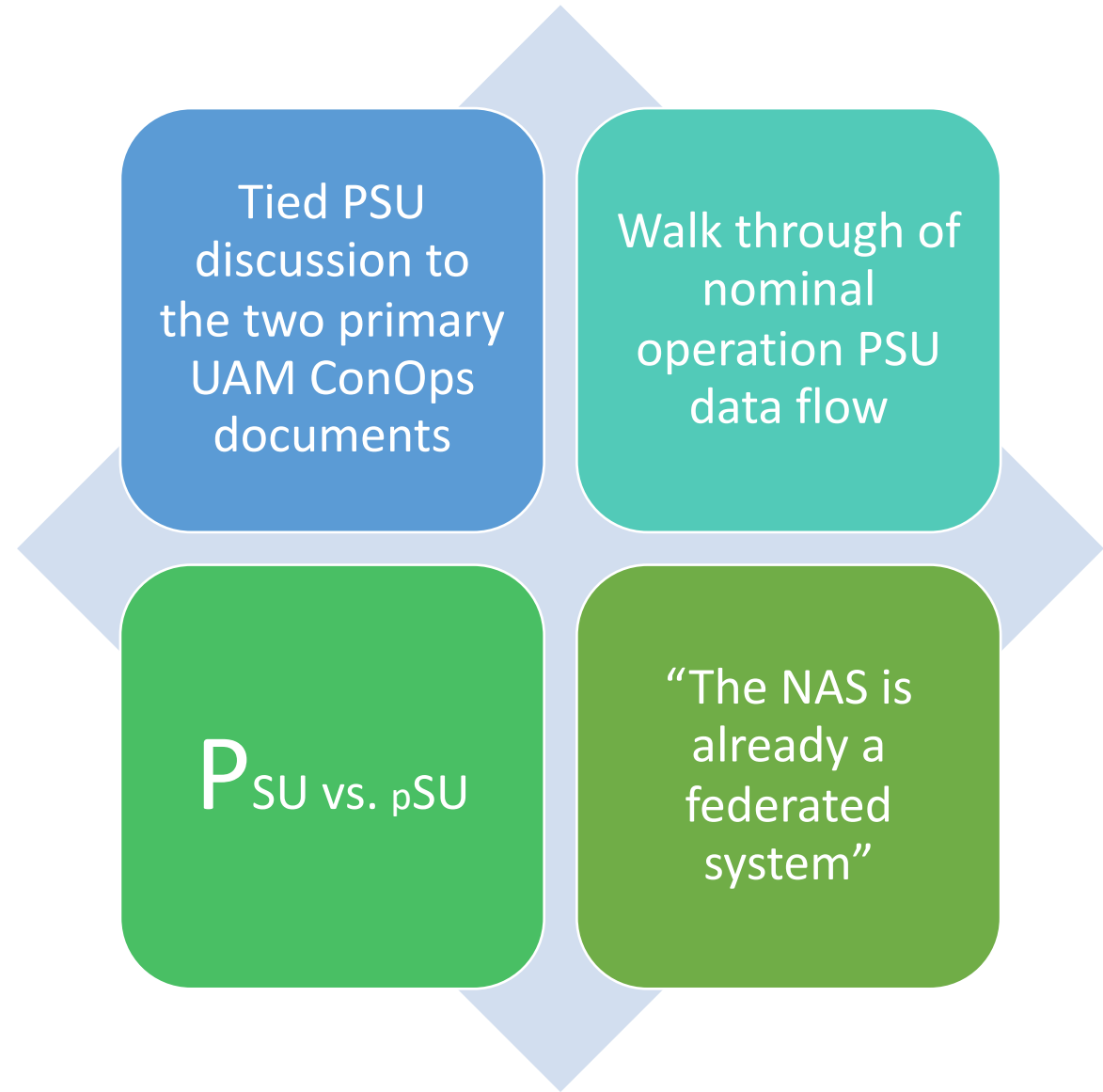
- A. Via radio voice communications
- B. With the PSU as an intermediary
- C. Not at all. ATC does not need to communicate any information to/from a UAM operator in nominal operation.
- D. Flag semaphores and aircraft marshalling signals

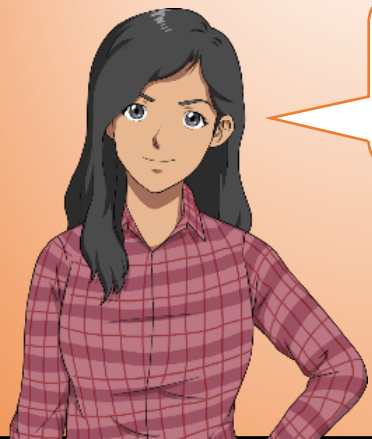
How does an organization become a recognized PSU?

- A. No formal government process, allow PSUs to self-certify and operators to choose the best one for their needs.
- B. Through some MOU process with the FAA.
- C. Via some qualification process other than MOU or Certification
- D. Via a Certification process
- E. Arm wrestling, biggest guy or gal from FAA vs biggest guy or gal from candidate PSU

Session 2

Review from last session

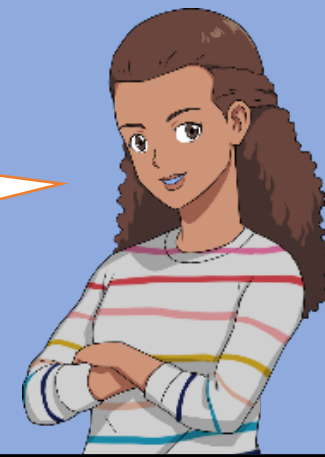




How you should react if you agree with the stuff Joey says...

I might say some things in today's meeting that are not necessarily community-vetted. Here is how you should take my statements today...

How you should react if Joey says things that seem off to you...



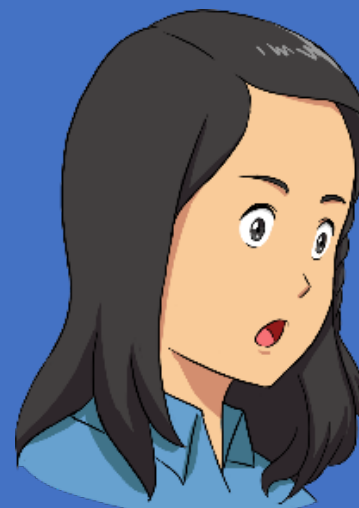
Everything Joey said was super correct!



Indeed! Very insightful!



Joey said some weird stuff today.



He was just playing devil's advocate to spur conversation! So wise!



Simple Questions from Last Session

- Will there be a FIMS? Yes. But it may have a different name and may supply data/services that are not the same as those in UTM.
- What is a USS? UAS Service Supplier, as defined in the UTM world
- Are there any identified PSUs out there? pSU maybe. PSU is for a future UML.
- Joey says Strategic deconfliction not equal to most safety critical; will you all discuss then what is the most safety critical way? I said that the word “primary” in discussing strategic deconfliction doesn’t equal/imply most safety critical.
- What is the role/interaction of PSU with a highly autonomous aircraft? What is the role of ATM with highly skilled pilots today? PSUs manage information flow about operations. On the scale of “management to control” they are more toward the management end. They help ensure that more tactical systems do not get overwhelmed. That is why ICAO has three layers for conflict management that still apply in a more autonomous environment like UAM.

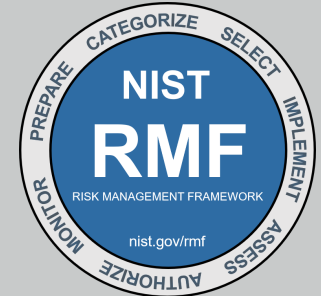
Less Simple Questions from Last Session

- Who owns or provides DSS? Open question. In UTM it is likely that industry provides an “accepted” implementation of DSS that meets some blessed standard. Likely the same in UAM, perhaps even the same DSS. PSUs need to prove the “SLA” of the DSS is true and appropriate and there needs to be mitigations to loss of DSS. Many open questions.
- Discuss how COP will be developed and role of PSU? The PSU Network is the primary source for UAM operational data. USS Network is primary source of UTM operational data. FIMS/FAA likely primary source for airspace data. An authorized entity like a PSU or USS would have access to these sources at the appropriate fidelity and could develop a COP along with other data sources. I don’t know that there is some unified GUI that is blessed as THE COP, if that is the question.
- Who funds the PSUs and to whom will they be accountable? A PSU would self-fund based on their individual business cases. They are Certified by the FAA (at least according to our poll last time) and are accountable for the set of services they provide. As a community, we need an overarching RACI chart with rows for specific features/actions.
- What is the underlying network of system that the PSUs will operate on? Current wisdom says the Internet is the underlying network. That could change, but I have not seen alternatives proposed or requested.
- What is the plan for cybersecurity architecture for PSUs? Ongoing work growing out of UTM. NASA UTM Project put in significant effort in this direction, FAA UPP2 pushed the envelope further. UAM building on those efforts. Some acronyms: NIST, IATF, IETF



The slide most likely to get Joey in trouble with every single person and organization.

While the concepts of external systems and external service providers are not new, the current pervasiveness and frequency of their invocation can present organizations with significant, new challenges. There are instances where system elements, subsystems, or perhaps the entire system may be outside of the direct control of the organization that authorizes its operation.... FISMA and OMB policy require external providers that process, store, or transmit federal information or operate information systems on behalf of the federal government to meet the same security and privacy requirements as federal agencies. Federal security and privacy requirements also apply to external systems storing, processing, or transmitting federal information and any services provided by or associated with the external system.



D.11.3 Air Transportation Information Type

Air Transportation involves the activities related to the safe passage of passengers or goods through the air. It also includes command and control activities related to the safe movement of aircraft through all phases of flight for commercial and military operations. Note: The protection of air transportation from deliberate attack is included in the Transportation Security information type under the Homeland Security mission area. The general recommended security categorization for the air transportation information type is as follows:

Security Category = {(confidentiality, Low), (integrity, Low), (availability, Low)}

Above quote from From NIST 800-37, Left quote from 800-60v2

Joey's view of a PSU "Authority to Operate" likely harsher than industry's view and maybe harsher than the FAA's as well.

Since PSUs are managing the airspace, and ATM is a function of the FAA that the FAA is not abdicating, the PSUs may be considered acting on behalf of the FAA.

All of this equates to PSUs likely needing to follow the NIST Risk Management Framework to become authorized to operate.

[NIST RMF](#), [NASA USS Auth](#), [NASA USS Spec](#), [UPP2 Report](#), [UPP2 Security Analysis](#), [NIST 800-60v2](#)

Operator vs. Pilot

- Operator is the person or organization responsible for operational control of the flight
 - See [8900.1](#) for more info
 - Examples of operational control functions include: preflight planning, canceling a flight due to unsafe conditions, crewmember qualifications, designating PIC/SIC, providing other personnel for dispatch, etc.
 - Often called a “Fleet Operator” in the AAM domain.
- Pilot in Command – The pilot responsible for the operation and safety of an aircraft during flight time. ([pilot/controller glossary](#), [CFR 14 91.3](#))
- In UAM (and UTM) it is the Operator is the one that communicates with the PSU (USS). Communications are assumed to be primarily digital in nature, facilitated by machine-to-machine protocols.
- *Note: a “Vertiport Operator” is yet another role with responsibilities related to managing a vertiport. This is distinct from the “Operator” above.*

Use Case Assumptions

PLEASE NOTE: These are assumptions for the Use Case... not 'universal' assumptions about UAM.

Airspace construct and procedures exist to facilitate UAM operations.

UAM Operations are in and around DFW.

Up to 100 operations may be airborne at any given time.

Several vertiports are in operation to support UAM.

UAM Operations are capable of VFR flight in uncontrolled airspace.

UAM Operations stay within defined UAM airspace structures when in controlled airspace.

Nothing bad or unexpected happens during the lifespan of our use cases today. Bad stuff is bad.

Use Case

Three operators using three different PSUs plan to depart an operation from three different vertiports.

No other operations departing those vertiports.

All three operations plan to land at the same vertiport at roughly the same time.

The operations are all reasonably spaced and sequenced in the air leading to the vertiport (“en route”) per protocols (rules, CBRs, etc.) and using structured procedural routes to the vertiport.

The vertiport has 5 FATOs (final approach and takeoff areas) available and the vertiport operator has provided a landing reservation to each of the operations at 3 different FATO.

Which description best fits my current role?

- A. FAA civil servant or contractor
- B. NASA civil servant or contractor
- C. Other government employee
- D. Industry stakeholder for UAM
- E. General Interest

When planning to operate in a designated UAM airspace construct, a UAM Operator _____ use a PSU to announce intent to other stakeholders.

- A. SHALL
- B. SHOULD
- C. MAY
- D. WOULD NEVER EVER

When expected capacity for 2 hours from now changes at a Vertiport, a UAM Operator with a planned landing around that time at that Vertiport would PRIMARILY learn about this change via _____.

- A. its PSU
- B. the Vertiport Operator
- C. the FAA
- D. a status announcement on a webpage
- E. some other means than those noted above

If an operator submits an operation plan to its PSU that conflicts with at least one other announced UAM operation, that PSU _____ assist its operator in deconflicting that operation plan.

- A. SHALL
- B. SHOULD
- C. MAY
- D. WOULD NEVER EVER

Where/when does the responsibility for conformance monitoring of an operation's 4D Trajectory end for a PSU?

- A. At the arrival fix for a vertiport
- B. At the FATO
- C. At the gate
- D. Some other location/time

If an Operator submits an operation plan (Operation Alpha) to its PSU that conflicts with at least one other announced UAM operation (Operation Beta) but meets all other UAM operational requirements, that PSU _____ announce Operation Alpha to the PSU Network as provided by the Operator. *(bonus discussion question: what should this PSU tell the operator of Operation Alpha?)*

- A. SHALL
- B. SHOULD
- C. MAY
- D. SHALL NOT

If a UAM Operation needs to perform a tactical maneuver for safety purposes, the Operator must communicate the maneuver to the PSU

_____.

- A. prior to executing that maneuver to ensure deconfliction with all elements of the airspace
- B. as soon as practical
- C. as soon as practical and only if the maneuver takes the operation out of conformance with its plan
- D. after landing and closing out the plan
- E. never

A PSU _____ subscribe to other services (like weather, route optimization, battery prognostics, etc.) to provide value to its supported Operators.

- A. SHALL
- B. SHOULD
- C. MAY
- D. SHALL NOT

A RADAR surveillance service is a PSU.

- A. TRUE
- B. FALSE
- C. EVERYTHING EVERYWHERE IS A PSU

A Vertiport Operator is a PSU.

- A. TRUE
- B. FALSE
- C. EVERYTHING EVERYWHERE IS A PSU

If an approach fix to a vertiport has an in-trail spacing requirement, for a given operation it is the responsibility of the _____ to ensure that in-trail spacing requirement is met.

- A. Individual Operator
- B. Vertiport Operator
- C. Supporting PSU
- D. Vehicle automation
- E. Pilot-in-command (remote or otherwise)

An Operator _____ implement and provide its own PSU services.

- A. SHALL
- B. SHOULD
- C. MAY
- D. CANNOT

Non-exhaustive, non-official list. Will evolve with rules and standards.

Other services will be necessary to fully enable UAM operations, but that doesn't mean they are PSU services.

Key PSU services/features



Checking plan versus airspace constructs and accepted rules/protocols

Is plan using airspace constructs appropriately?
Any other required checks performed by operator (wx, etc.)?



Bridging operator-FAA strategic, digital communications

Airspace authorizations
Notification of off-nominal events or encroachment on controlled, non-UAM airspace



Sharing intent and state with other PSUs per CBRs

Ensure that initially shared plans are deconflicted
Provide position/state reports to other PSUs as appropriate



Monitoring conformance of an operation with its plan

Alert operator in case of non-conformance or potentially nearly non-conformant
Update PSU Network in cases of contingency

Key PSU services/features

Proposal: A entity CANNOT be a PSU unless it meets the following draft requirements



Checking plan versus airspace constructs and accepted rules/protocols

A PSU SHALL ensure that operation plans that it supports appropriately use airspace constructs

A PSU MAY assist an operator in meeting the operators non-PSU related requirements



Bridging operator-FAA strategic, digital communications

A PSU SHALL notify the FAA of an authorized operation

A PSU SHALL notify the FAA of off-nominal events or encroachment on controlled, non-UAM airspace for operations that the PSU is supporting



Sharing intent and state with other PSUs per CBRs

A PSU SHALL ensure that operation plans are appropriate deconflicted per “rules” prior to sharing with the PSU Network

A PSU SHALL share operation state with the PSU Network per “rules”



Monitoring conformance of an operation with its plan

A PSU SHALL alert an operator that it is supporting when that operator’s operation is non-conforming with its plan

A PSU SHALL alert the PSU Network when an operation it is supporting enters a Contingent state

Detailed Use Case

Use Case ConOps Driver

- *The interactions between UAM aerodromes and PSUs is a key research area that will define the domains of the UAM aerodrome operator and PSUs with respect to scheduling and sequencing operations into and departing from aerodromes, how contingencies requiring alternate landing locations are managed, and the level of trajectory precision required to support operations on high-density routes between high-demand aerodromes.*

Use Case ConOps Driver

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Use Case

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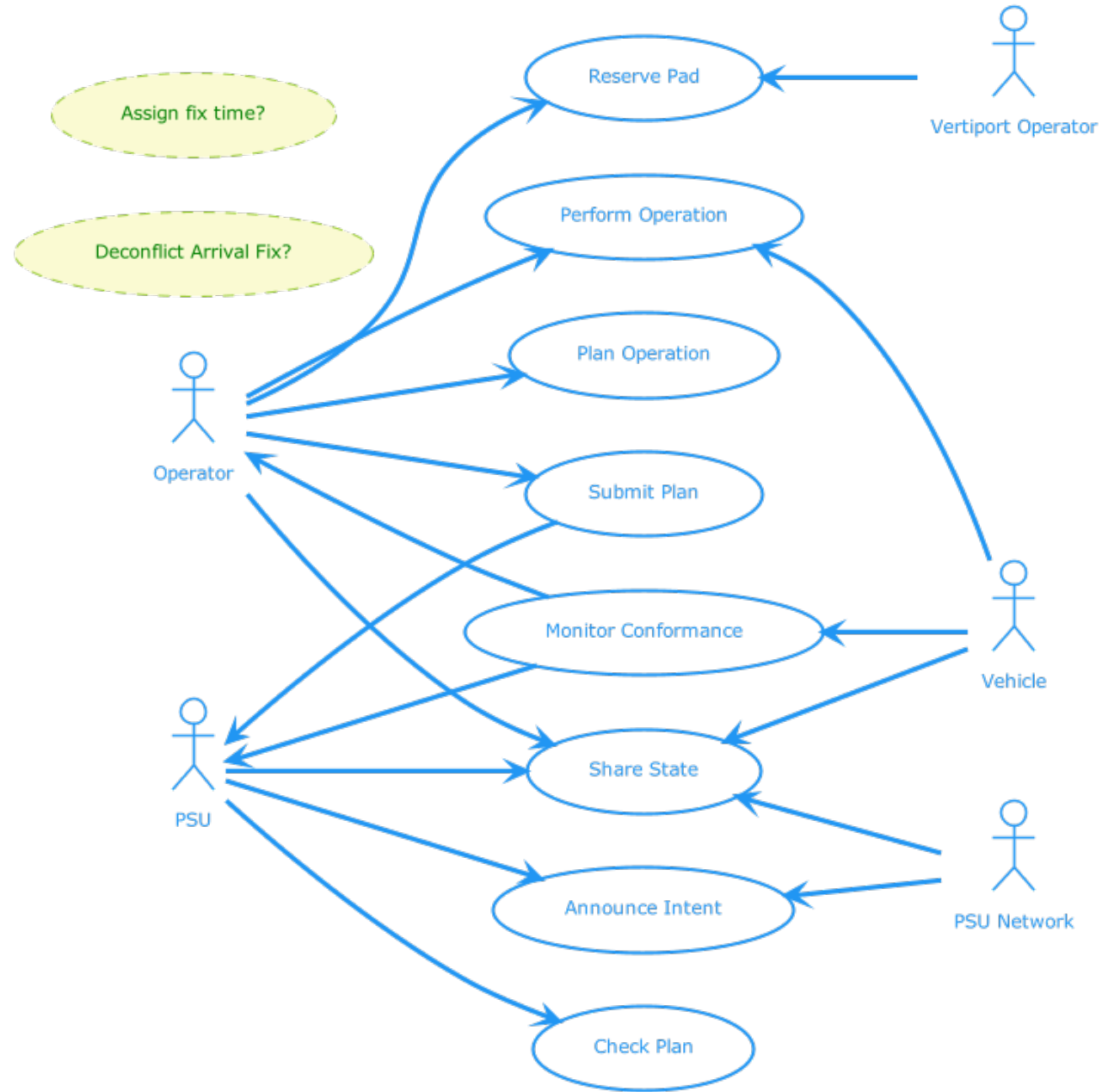
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Nominal Operations with Vertiport





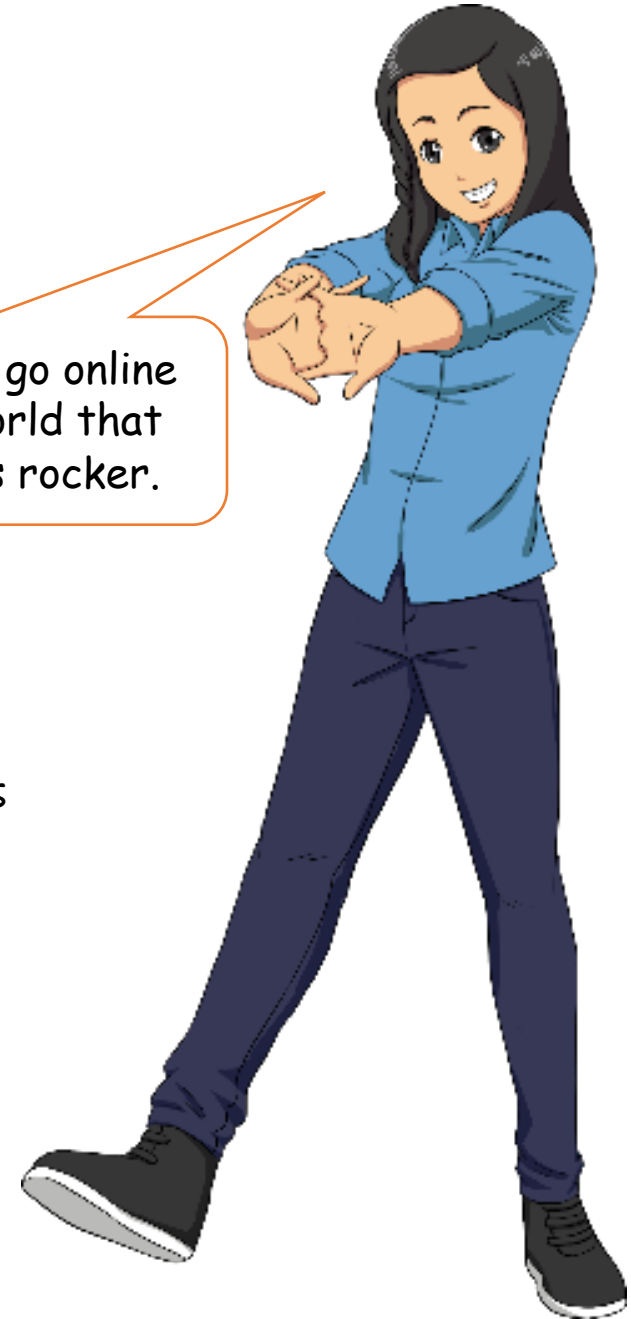
Questions

- Whose task is it to ensure orderly and safe landing sequence at the vertiport?
- When is this task supposed to be completed? Or is it continuous? If continuous does the responsibility shift between entities at any time?
- When does conformance monitoring via PSU end? Approach fix? Wheels-down? On the pad? Later?
- Can this task of deconflicting three operations to the same vertiport be effectively managed by the operator, PSU, and/or vertiport operator or is some other entity/service needed?
- Does voice comm play a role in the management of this nominal scenario? If so, who is talking to whom and when?



That was super great, I totally get PSUs now. I think we may have completely solved UAM!

Whelp, time to go online and tell the world that Joey is off his rocker.



Thanks! I'm assuming everyone is in one of these two camps. Feel free to reach out:

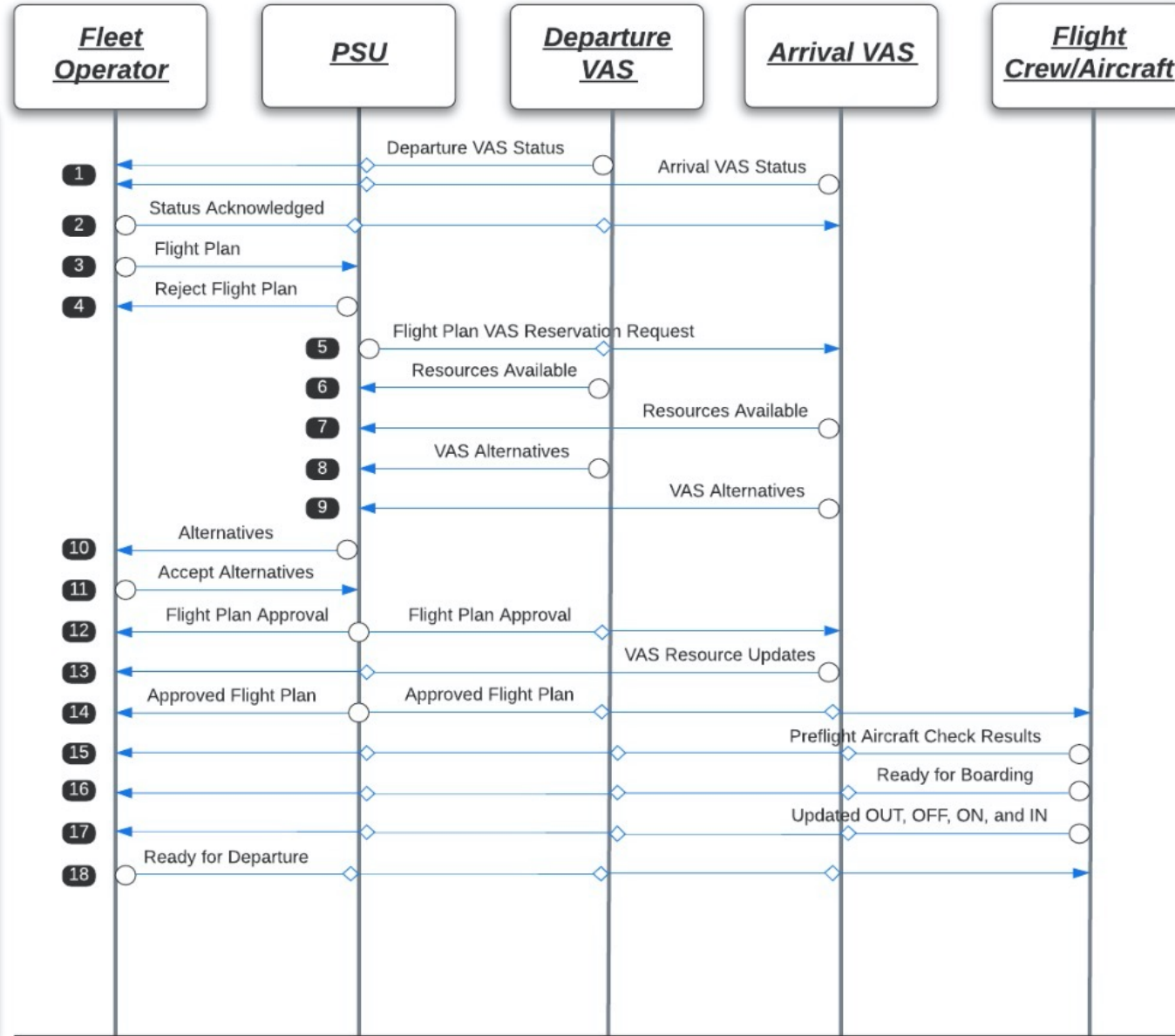
Joseph.L.Rios@nasa.gov

Backup

Legend:

- Starting Point
- ◇ Info. Sharing
- ➡ End Point
- # Sequence Step

1. VAS broadcast status and resource availability
2. Acknowledge receipt of VAS message
3. Fleet Operator files Flight Plan
4. PSU checks Flight Plan and rejects if not correct
5. PSU submits flight plan to Departure and Arrival VAS
6. Departure VAS Resources Available
7. Arrival VAS Resources available
8. Departure VAS Resources NOT Available, sends alternatives
9. Arrival VAS Resources NOT Available, sends alternatives
10. PSU sends alternatives to Fleet Operator
11. Fleet Operator accepts alternatives
12. PSU accepts Flight Plan
13. VAS secures reservations and updates resource availability
14. PSU distributes Flight Plan
15. Aircraft preflight check results distributed
16. Aircraft final check successful, ready for boarding released
17. Aircraft updates OUT, OFF, ON, IN times
18. Fleet Operator dispatches aircraft ready for departure



UTM Conflict Management Model v2.0		Strategic Separation	Tactical Separation		
		Strategic Conflict Management	Separation Provision		Collision Avoidance
UTM	USS Function	Flight Awareness Service	Conformance Monitoring Service		
		Strategic Deconfliction Service	Dynamic Rerouting Service		
	SDSP or USS Function	Operation Planning Service	Conflict Advisory and Alerting Service		
		Flight Notification Service	Surveillance Service		
	UAS Operator / UAS Function	Operation Planning	Ground Surveillance		Visibility / Audibility Enhancements
			Position Broadcast / Remote Identification	Detect and Avoid	Collision Avoidance
			Geographic Flight Containment		Obstacle Avoidance

Airspace Hazards

Airborne Hazards

Ground Hazards

From: [UAS Traffic Management Conflict Management Model](#)